BENCHMARKING NEURAL NETWORK ROBUSTNESS TO COMMON CORRUPTIONS AND PERTURBATIONS

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arXiv:1903.12261v1 [cs.LG] 28 Mar 2019

ABSTRACT

In this paper we establish rigorous benchmarks for image classifier robustness. Our first benchmark, IMAGENET-C, standardizes and expands the corruption robustness topic, while showing which classifiers are preferable in safety-critical applications. Then we propose a new dataset called IMAGENET-P which enables researchers to benchmark a classifier's robustness to common perturbations. Unlike recent robustness research, this benchmark evaluates performance on common corruptions and perturbations not worst-case adversarial perturbations. We find that there are negligible changes in relative corruption robustness from AlexNet classifiers to ResNet classifiers. Afterward we discover ways to enhance corruption and perturbation robustness. We even find that a bypassed adversarial defense provides substantial common perturbation robustness. Together our benchmarks may aid future work toward networks that robustly generalize.

1 INTRODUCTION

The human vision system is robust in ways that existing computer vision systems are not (Recht et al., 2018; Azulay & Weiss, 2018). Unlike current deep learning classifiers (Krizhevsky et al., 2012; He et al., 2015; Xie et al., 2016), the human vision system is not fooled by small changes in query images. Humans are also not confused by many forms of corruption such as snow, blur, pixelation, and novel combinations of these. Humans can even deal with abstract changes in structure and style. Achieving these kinds of robustness is an important goal for computer vision and machine learning. It is also essential for creating deep learning systems that can be deployed in safety-critical applications.

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Most work on robustness in deep learning methods for vision has focused on the important challenges of robustness to adversarial examples (Szegedy et al., 2014; Carlini & Wagner, 2017; 2016), unknown unknowns (Hendrycks et al., 2019; Hendrycks & Gimpel, 2017b; Liu et al., 2018), and model or data poisoning (Steinhardt et al., 2017; Hendrycks et al., 2018). In contrast, we develop and validate datasets for two other forms of robustness. Specifically, we introduce the IMAGETNET-C dataset for input *corruption robustness* and the IMAGENET-P dataset for input *perturbation robustness*.

To create IMAGENET-C, we introduce a set of 75 common visual corruptions and apply them to the ImageNet object recognition challenge (Deng et al., 2009). We hope that this will serve as a general dataset for benchmarking robustness to image corruptions and prevent methodological problems such as moving goal posts and result cherry picking. We evaluate the performance of current deep learning systems and show that there is wide room for improvement on IMAGENET-C. We also introduce a total of three methods and architectures that improve corruption robustness without losing accuracy.

To create IMAGENET-P, we introduce a set of perturbed or subtly differing ImageNet images. Using metrics we propose, we measure the stability of the network's predictions on these perturbed images. Although these perturbations are not chosen by an adversary, currently existing networks exhibit surprising instability on common perturbations. Then we then demonstrate that approaches which enhance corruption robustness can also improve perturbation robustness. For example, some recent architectures can greatly improve both types of robustness. More, we show that the Adversarial Logit Pairing ℓ_∞ adversarial example defense can yield substantial robustness gains on diverse and common perturbations. By defining and benchmarking perturbation and corruption robustness, we facilitate research that can be overcome by future networks which do not rely on spurious correlations or cues inessential to the object's class.

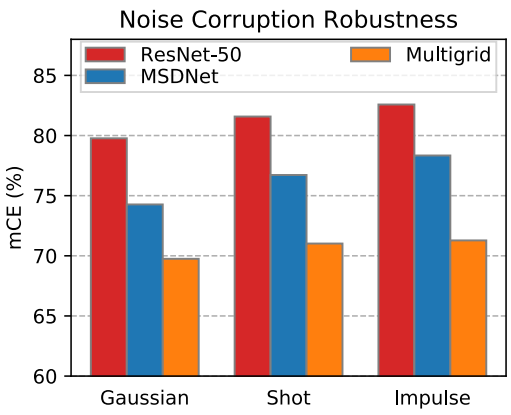


Figure 5: Architectures such as Multigrid networks and DenseNets resist noise corruptions more effectively than ResNets.

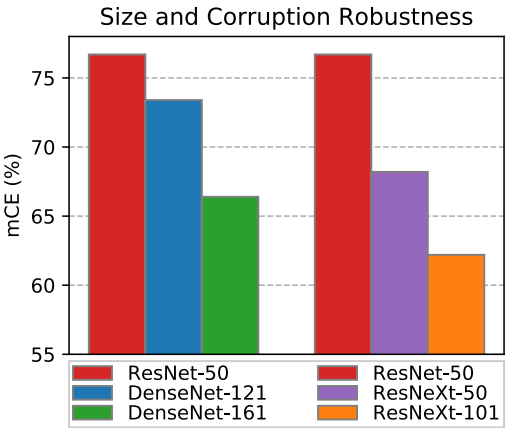


Figure 6: Larger feature aggregating networks achieve robustness gains that substantially outpace their accuracy gains.

Both model size and feature aggregation results are summarized in Figure 6. Consequently, future models with even more depth, width, and feature aggregation may attain further corruption robustness.

Feature aggregation and their larger counterparts similarly improve perturbation robustness. While a ResNet-50 has a 58.0% mFR and a 78.3% mT5D, a DenseNet-121 obtains a 56.4% mFR and 76.8% mT5D, and a ResNeXt-50 does even better with a 52.4% mFR and a 74.2% mT5D. Reflecting the corruption robustness findings further, the larger DenseNet-161 has a 46.9% mFR and 69.5% mT5D, while the ResNeXt-101 has a 43.2% mFR and 65.9% mT5D. Thus in two senses feature aggregating networks and their larger versions markedly enhance robustness.

Stylized ImageNet. Geirhos et al. (2019) propose a novel data augmentation scheme where ImageNet images are stylized with style transfer. The intent is that classifiers trained on stylized images will rely less on textural cues for classification. When a ResNet-50 is trained on typical ImageNet images and stylized ImageNet images, the resulting model has an mCE of 69.3%, down from 76.7%.

Adversarial Logit Pairing. ALP is an adversarial example defense for large-scale image classifiers (Kannan et al., 2018). Like nearly all other adversarial defenses, ALP was bypassed and has unclear value as an adversarial defense going forward (Engstrom et al., 2018), yet this is not a decisive reason dismiss it. ALP provides significant perturbation robustness even though it does not provide much adversarial perturbation robustness against all adversaries. Although ALP was designed to increase robustness to small gradient perturbations, it markedly improves robustness to all sorts of noise, blur, weather, and digital IMAGENET-P perturbations—methods generalizing this well is a rarity. In point of fact, a publicly available Tiny ImageNet ResNet-50 model fine-tuned with ALP has a 41% and 40% relative decrease in the mFP and mT5D on TINY IMAGENET-P, respectively. ALP's success in enhancing common perturbation robustness and its modest utility for adversarial perturbation robustness highlights that the interplay between these problems should be better understood.

6 CONCLUSION

In this paper, we introduced what are to our knowledge the first comprehensive benchmarks for corruption and perturbation robustness. This was made possible by introducing two new datasets, IMAGENET-C and IMAGENET-P. The first of which showed that many years of architectural advancements corresponded to minuscule changes in relative corruption robustness. Therefore benchmarking and improving robustness deserves attention, especially as top-1 clean ImageNet accuracy nears its ceiling. We also saw that classifiers exhibit unexpected instability on simple perturbations. Thereafter we found that methods such as histogram equalization, multiscale architectures, and larger feature-aggregating models improve corruption robustness. These larger models also improve perturbation robustness. However, we found that even greater perturbation robustness can come from an adversarial defense designed for adversarial ℓ_∞ perturbations, indicating a surprising interaction between adversarial and common perturbation robustness. In this work, we found several methods to increase robustness, introduced novel experiments and metrics, and created new datasets for the rigorous study of model robustness, a pressing necessity as models are unleashed into safety-critical real-world settings.

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